

Purpose

To provide a traceable record of the work completed to bring this product toward fruition. The document captures both successes and failures so that future engineers can avoid repeating the same mistakes. It can also serve as a foundation for final product documentation.

Goal

This project models a product purchasing system. A client machine accepts customer orders and stores them in a SQL database. The database then notifies the email server to send an invoice receipt to the customer's email address.

Assignment Summary

This assignment introduced the principles and history of cloud computing. Because I had limited prior exposure to the subject, the work was challenging; however, I successfully created and launched three instances. I gained practical experience with virtual private clouds (VPCs), Amazon EC2, security groups, subnets, and internet gateways. I also encountered configuration issues, including initially omitting public IP addresses, which prevented Remote Desktop connections. A remaining requirement is to enable communication among the three instances; that functionality was not operational at the time of writing.

Claim

I created three Amazon EC2 instances in AWS:

- Microsoft Windows Server 2025 application server
- Microsoft SQL Server 2022
- Ubuntu email server

Assumptions

1. Following the AI-generated instructions step by step would result in fully functional instances.

Evidence

Screenshots document the VPC, EC2 instances, security groups, subnet, and related AWS resources.

Validation

I confirmed that each instance functions independently; however, the instances currently fail to communicate with one another.

AI Assistance

I used Microsoft Copilot to provide detailed, step-by-step guidance for provisioning the instances and configuring IP addressing, security settings, and network resources. I also used Copilot to assist with reformatting this document.

Ownership

I am responsible for reviewing the AI-generated guidance, implementing and validating the configuration, documenting results, and escalating unresolved technical or security risks.

Open Questions and Escalation

1. How can the instances be configured to communicate reliably? Without this capability, the project cannot proceed.
2. Does the current security configuration adequately protect financial information?
3. How will customers' purchasing data and other personal information be safeguarded?

Escalation requirement: These questions should be resolved before the project advances to the next phase.

Documentation Included

- Prompts submitted to Microsoft Copilot for project assistance
- Screenshots of relevant AWS console resources
- AI-generated instructions and output
- Known uncertainties and unresolved issues